Tracking Semantic Change in Technology-Related Words Using Temporal Word Embeddings

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News Category Dataset · Word2Vec Skip-gram · Orthogonal Procrustes Alignment · Streamlit Demo

Motivation

Word meanings change over time because technology, society, and media evolve. Static word embeddings represent each word with a single vector and cannot directly capture diachronic semantic change.

This project uses temporal Word2Vec embeddings to analyze how selected words changed their contexts between the **2010s** and the **2020s** in news texts.

Dataset and Experimental Setup

Dataset

News Category Dataset

News headlines and short descriptions were merged into one text field. Publication dates were converted into years.

Period	Years	Documents
2010s	2012–2019	19,249
2020s	2020–2022	5,518

Target and Control Words

Main target words:

model · network · platform · virus

Additional target words:

web · stream

Control words:

water · city · winter · animal · family · school

Model Setup

Parameter	Value
Model	Word2Vec Skip-gram
vector_size	100
window	5
min_count	5
epochs	10
seed	42
alignment	Orthogonal Procrustes

Methodology

Pipeline

Corpus → Preprocessing → Time slicing → Word2Vec training → Procrustes alignment → Cosine shift → Nearest neighbors

Separate embeddings were trained for each period. Since independently trained vector spaces are not directly comparable, the spaces were aligned using Orthogonal Procrustes. Semantic shift was measured as cosine distance from the base period.

Best Shift Scores

0.54	0.53	0.53	0.35
model	network	platform	virus

Alignment Note

Target and control words were excluded from the anchor vocabulary during alignment to avoid artificially reducing the measured semantic shift.

Experimental Results

Main Quantitative Results

Word	Cosine distance	Interpretation
model	0.54	Strong shift
network	0.53	Strong shift
platform	0.53	Strong shift
virus	0.35	Moderate shift

Figure 1. Semantic Shift from Base Period

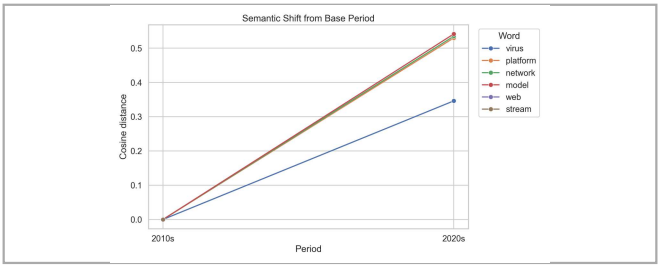
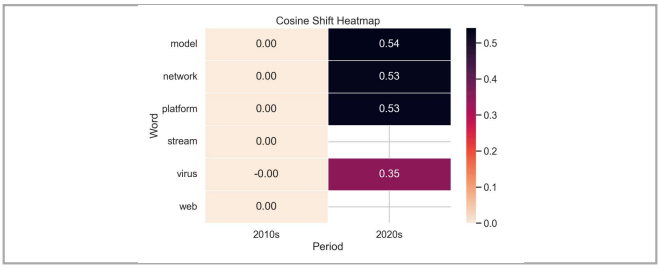


Figure 2. Cosine Shift Heatmap

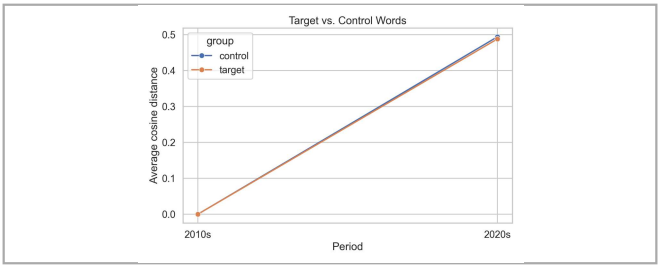


Interpretation and Discussion

- **model** showed the strongest semantic movement.
- **network** shifted toward media, political, business, and communication contexts.
- **platform** shifted toward digital, media, social, and communication contexts.
- **virus** shifted toward disease and pandemic-related contexts in the 2020s.

Target vs. Control

The target-control difference was modest. Therefore, the main conclusion is based on strong individual word shifts rather than a large group-level effect.



Interactive Demo Example

network → 0.535 → Strong semantic shift detected

The Streamlit app allows users to enter a word and view cosine shift, nearest neighbors by period, and shift strength interpretation.

```
streamlit run app.py
```